

What can be claimed about a corpus of seven thousand signs: null tests, calibration and confounders in Linear A, with Etruscan as control

Alberto Acedo Biome Makers Inc. Draft v0.2, 5 September 2026. Sections 6 and 7 will be revised in the light of the external evaluation of the reading they discuss.

Abstract

Small undeciphered corpora invite two opposite errors: readings obtained from sound alone, which any large lexicon can supply, and agnosticism that treats every statement as untestable. This paper describes a procedure that sits between the two and applies it to Linear A (7,000 signs, 1,719 documents, 799 distinct sign-groups): every distributional claim is tested against a permutation null that preserves what must be preserved; the instruments are calibrated beforehand on Etruscan, where the answers are known; the confounders of site, period, genre and scribal hand are separated where the data allow and declared inseparable where they do not; and negative results are reported with their numbers. On Linear A the procedure fixes a map of what the corpus can support: the language is not Greek even when read with Greek values; the phonotactic distance between the accounting archives and the cult inscriptions is as large as the distance to Greek and its causes cannot be separated; twenty names of the Haghia Triada archive survive in Knossian Linear B under an explicit rule; three suffixes and one prefix have measurable distributions; vocabulary sharing at Haghia Triada is explained by scribal hand rather than by document type; sound-based matching does not discriminate among candidate languages; and two-syllable form matches carry no signal in either Aegean script. The same procedure produced, and then corrected, one lexical proposal, reported elsewhere.

Keywords: Linear A, Etruscan, permutation tests, null models, calibration, confounders, undeciphered scripts, computational epigraphy.

1. The need

Linear A has some seven thousand signs distributed over 1,719 documents, most of them accounting tablets from Haghia Triada and Khania, with no bilingual and no known linguistic relative. The attempts at reading, from Gordon to Georgiev and from Palmer to Finkelberg, have followed the same path: assign to the signs the values of Linear B, search a known language for similar words, and find them, because a syllabary of ninety signs and words of two or three syllables produce matches with any large lexicon (Duhoux 1989). The recent computational study by Nepal and Perono Cacciafoco (2024) acknowledges three limitations, which are the ones this paper addresses: not filtering chance matches, not distinguishing loan from chance, and not measuring the power of a small corpus. The editors of the first collective volume devoted to the script (Petrakis and Steele 2025) add an observation taken here as the starting point: the published corpus is equivalent to that of a single scribe at Pylos, and there is no reason to suppose that a “critical point” of quantity exists beyond which results become reliable; the problem is not only size but the absence of a term of comparison.

What is proposed here is not a method of reading but a procedure for fixing the map of what can be claimed. Its components are not new: the distributional analysis is that of Duhoux and Davis, the fixed-margin permutation nulls come from community ecology (Gotelli 2000; Strona et al. 2014), calibration on a corpus with known answers is standard practice for any instrument. What had not been done in this field is to combine them with discipline and to publish the negatives with their numbers, and that is what follows.

2. Data and procedure

2.1 Corpus

GORILA transliterations (Godart and Olivier 1976-1985) in the machine-readable form of the LinearA Explorer (Hogan 2022), which preserves subscripts, site, support, archaeological context, the hand attributions of GORILA V and Younger's commentaries (2024b) with the seal data. Sign verification against the SigLA dataset (Salgarella and Castellan 2020). For Linear B, the same author's edition of the standard editions (5,889 documents, 5,597 sign-groups), with Olivier's Knossos hands. For Etruscan, the Larth corpus (Vico) with 7,139 records and the ETP corpus (429 texts with metadata). A known defect of the Explorer data, the support label of the Zakros tablets, was corrected from SigLA; one ligature in its transliteration (*304+PA-KU-PA on HT Wa 1020) turned out to be two consecutive signs according to SigLA, and is recorded here as a warning about derived transliterations.

2.2 Units and measures

The units are sign-groups (799 types, 1,117 tokens on tablets with two or more groups; 76% singletons), documents as sets of groups, and profiles of initial syllables and of consonantal skeletons. The measures are the usual ones: Jensen-Shannon distance between profiles (equation 1), Jaccard index between documents (equation 2), Mantel correlation between distance matrices (equation 3), mutual information between positions (equation 4), and the curveball algorithm for fixed-margin nulls on presence matrices (equation 5, Strona et al. 2014). Each is stated in the appendix with its source.

2.3 Nulls

Each claim carries the null that preserves what the claim does not attempt to explain: for co-occurrence between documents, permutations that preserve the number of groups per document and the number of documents per group (curveball); for morphology, permutation of the affixed syllable among all words with their lengths preserved; for positional classes, permutation of labels; for confounders, permutation within strata (type label within hand, or hand within type). The threshold was set at $p < 0.01$ with one thousand permutations for new claims and $p < 0.05$ for replications; multiple comparisons are corrected by Bonferroni when fewer than ten and by Benjamini-Hochberg when more.

2.4 Power

Before each test the minimum detectable effect with the available n was estimated. Two figures govern the rest: the floor of distance between two samples of the same corpus at 42 types is

0.198 (Jensen-Shannon on consonantal skeletons), so that no smaller difference is detectable at that size; and in Etruscan the sibilant test has 68 usable pairs, which is the order of magnitude of the positional tests in Linear A. A test whose expected effect falls below the floor is declared undecidable and is not run.

3. Calibration on Etruscan

Etruscan is the appropriate control: a read alphabet, a non-Indo-European language half understood, a corpus of the size of Linear A with metadata of city and period, and three established grammatical or sociolinguistic facts that the procedure must recover without being told.

Genitive before clan. Before the word clan, “son”, the father’s name stands in the genitive (-s, -l). The positional test, applied blind, finds the endings -s and -l in 58% of the words preceding clan (and sec, “daughter”, and puia, “wife”) against 26% in the rest of the corpus ($p = 0.002$ by permutation). The instrument sees inflection where it exists.

Syncope as change over time. The loss of internal vowels is dated around 500 BC. The syncope index is 0.28 higher in late texts than in early ones, and the effect survives control by city ($p < 0.002$ within city). The instrument sees time where it exists.

Sibilants as geographic variation. The graphic distinction of sibilants is a northern trait. The north/south ratio is 0.6 against 0.1 ($p = 0.72$ within city): the effect is one of city, not of period. The instrument does not confuse geography with time when the metadata allow them to be separated.

The calibration fixes what may be expected from the same apparatus on Linear A: it will recover morphology and stratification if they exist and if metadata allow the confounders to be separated; where they do not, it will say “undecidable”, and that answer must be accepted.

4. Results with nulls on Linear A

The subsections that follow report the claims that survived the procedure of 2.4. Each carries its null and, where applicable, its correction; those that did not survive are in section 5 or in the register accompanying this paper.

4.1 Not Greek, not even in its own syllabary

The Jensen-Shannon distance between the skeleton profiles of Linear A and Linear B, read with the same values, is 0.104, against a floor of 0.020 between two halves of A and 0.092 for a null in which the sign inventory of B is shuffled: A is further from Mycenaean Greek than scrambled Mycenaean Greek. A converter from alphabetic Greek into the syllabary recovers 36% of whole words and 63% of roots (47% and 68% with the digamma and labiovelar rule); the 85% threshold that identity of language would require is not reached. A screen of seven candidate languages by consonantal skeleton gives ratios from 1.46 (Hittite) to 1.68 (Hebrew) over each lexicon’s own null, with Greek at 1.47: none stands out, and the highest ratio belongs to the largest lexicon. Comparison by sound, whatever the language, does not discriminate.

4.2 Two registers, with inseparable causes

The distance between the accounting tablets of Haghia Triada (LM IB) and the cult inscriptions of the sanctuaries (MM III-LM IA) is 0.193, of the same order as the distance to Greek (0.194). The four possible causes (genre, site, language, period) travel together in the corpus and cannot be separated: Phaistos, the only period control without a change of genre, has 42 types and lies below the floor. Within Haghia Triada, where site is constant, document types (personnel, produce, aromatics, allocation) do not differ in syllable profile above the null (Jensen-Shannon at 80 syllables, all pairs $p > 0.18$, with free permutation and with permutation within hand). The difference between archives and sanctuaries is therefore one of genre and site, not of administrative type; whether it is also one of language the corpus cannot say, and so it is stated. The variants of the libation formula are explained by site, not by grammatical agreement (p from 0.011 to 0.16 when controlling for site); the prefix *ja-/a-* has frequencies of 7 per thousand at Haghia Triada against 140-240 per thousand in the sanctuaries, which makes it a feature of genre.

4.3 Names that survive at Knossos

With Younger's explicit rule $-U/-E \rightarrow -o$, twenty sign-groups from Haghia Triada of three or more syllables reappear in Knossian Linear B, against 8.1 expected under a bigram null ($p < 0.003$); at Pylos, 7 against 3.5 ($p = 0.06$); at Thebes, none; among the sanctuary words, 0 of 47. Survival is asymmetric (names, not cult vocabulary) and local (Knossos, not the mainland), which is consistent with a Mycenaeanised administration that retains Cretan onomastics. Two non-Greek theonyms of Knossos (*pa-de*, *a-ta-na*) appear in A (2 of 15, $p = 0.04$); three Pre-Greek names of B (*mi-ta*, *sa-sa-me*, *ki-ri-ta₂*) appear at Haghia Triada (3 of 20, $p = 0.09$), with the Greek controls at 0 of 6.

4.3 bis What happens to a Minoan name in Greek mouths

The pairs of the previous section allow the correspondence to be measured. In 46 pairs of A groups of three or more syllables whose Linear B counterpart is identical but for the vowel of the final syllable, Greek gives *-o* in 63% of cases, against a base rate of 44% among B finals; Minoan *-a* is never preserved (ten of ten become *-o*: *ta-ni-ka* $\rightarrow$ *ta-ni-ko*, *a-ta-na* $\rightarrow$ *a-ta-no*), and finals in *-e*, *-i* and *-u* give *-o* or *-a*. Younger's rule $-U/-E \rightarrow -o$ is a particular case of a more general one: Greek reassigned Minoan names to its *o-* and *a-*stems without respecting the Minoan final vowel, so that vowel was not a gender marker. The full alignment (946 pairs differing in one syllable against 610 for the shuffled-syllable null, maximum 692) carries signal, but each individual consonantal correspondence is too scarce to be asserted.

One negative deserves recording. Of the 782 sign-groups of the corpus, none ends in *-so* and three in *-to* (0.4%), whereas the Cretan toponyms as Greek writes them end mostly in *-so* and *-to* (Knossos, Tylissos, Amnisos; Phaistos, Lyktos). The same places, written in Linear A, end otherwise: *su-ki-ri-ta*, *ja-di-ki-tu*, *a-di-ki-te*, *se-to-i-ja*, *ki-da-ro*. The substrate layer in *-ssos* and *-nthos*, as Greek sees it, does not appear in the words the Minoans wrote: either it is a Greek

adaptation of Minoan finals in -ta, -te, -tu and -ja, or it belongs to another stratum. It is a datum for an old debate (Furnée 1972; Beekes 2010), obtained without reading.

4.4 Morphology placed, not read

Three suffixes form root/derivative pairs above the shuffled-syllable null: -ja (8 pairs against 3.3, $p = 0.02$), -te (7 against 3.3, $p = 0.03$), -se (5 against 1.5, $p = 0.02$); none survives Bonferroni over eleven tests, and this is reported. The prefix ka- forms five pairs against 1.7 ($p = 0.013$) and concentrates its distribution on the roundels (17% of roundel groups against 4% of tablet groups). In the root/-ja pairs the bare form is a heading and the derived form an entry with a quantity; taken over all groups in -ja, the concentration disappears. This is grammar placed by position, not read. The prefix i- (and its form i-na-) is the strongest of all: eight pairs against 2.5 for the shuffled-first-syllable null (maximum 7 in 300 permutations, $p < 0.003$) and two with i-na- against 0.1, with the pair i-da-ma-te / da-ma-te (the same divine name with and without the prefix at two sanctuaries) and with i-na-i-da / i-da on place-names; Younger had noted it at Phaistos as a possible marker of “to / at”, and the null confirms it. Other prefixes stay within chance (ja-, u-, ta-, pa-), so the list of productive affixes of Linear A is: i-, i-na-, ka-, a-, ki- and the suffix -ja. There is no word for “son of”; filiation, if expressed, goes by suffix.

4.5 Compartmented vocabulary, and who compartments it

At Haghia Triada the accounts with a total (ku-ro) have six monopoly words (groups that appear in no other document type; null 1.1, $p < 0.003$) and the allocations (sa-ra₂) three (null 0.4, $p = 0.007$). There is no measurable co-exclusion between frequent words in the whole corpus (zero pairs with expected ≥ 1 and observed 0), and the documents $\times$ words metacommunity with the curveball null has a single module (the grain list of HT 86 and 95). With the hand attributions of GORILA V (73 tablets, 21 hands), vocabulary sharing between tablets is explained by hand (effect 0.032, $p = 0.002$ controlling for type) and not by document type (0.025, $p = 0.21$ controlling for hand); the compartments are, at least in part, scribal portfolios. The claim must be calibrated, and an external classification allows it. Montecchi (2010) grouped the Haghia Triada tablets into classes and series on the basis of content, text arrangement, find-spot and, where possible, hand. Tested by label permutation over the 88 tablets she classifies, the grouping holds: pairs of the same series share 0.215 of their logograms against 0.039 for pairs of different series ($p < 0.0005$), and they also share more vocabulary (0.044 against 0.012, $p < 0.0005$), which is not trivial, since vocabulary is not one of her criteria. Nine of the twelve series with three or more tablets have internal cohesion far above their surroundings; the three that do not are those defined by the absence of an ideogram. Over the 44 tablets that have both a series and an attributed hand, the decisive result is the interaction: two tablets of the same series and the same hand share 0.216 of their vocabulary; same hand and different series, 0.052; same series and different hand, 0.015; neither, 0.018. Both factors are significant separately (series within hand 0.201, $p < 0.003$; hand within series 0.164, $p < 0.003$), but shared vocabulary concentrates almost entirely where both coincide. The correct reading is therefore neither that the scribe explains the vocabulary nor that the document type does, but that it is explained by the scribe working within a particular dossier; our earlier claim, made with a coarse typology of our own, is

superseded by this one. The claim must be calibrated: in the Linear B of Knossos (1,158 tablets with hand and series), vocabulary is explained by the editors' series (effect 0.020, $p < 0.001$) and not by hand, but the series is a classification made by content, so the effect is partly tautological; the typology used here for Haghia Triada is coarse and our own, and the fair comparison (Montecchi's 2010 classes for HT, or a content-independent typology for Knossos) is pending. What stands is that the scribe is a confounder no lexical analysis of A had measured, and that its weight depends on the typology against which it is measured.

4.5 bis Arithmetic as proof, and what it proves

The only proof this corpus admits without external comparison is the accounting one, and it is what secured KU-RO as "total". With an entry table built for all the Haghia Triada tablets (word, logogram, quantity, sections separated by headings and totals), 25 totals have two or more quantities in their section and 8 add up exactly; those that do not are explained by lacunae, by fractions that the Explorer transliteration renders as integers (HT 13: 131 against $130 \frac{1}{2}$) or by section boundaries the parser cuts wrongly. It is a replication of Duhoux's result with a systematic list of where GORILA must be consulted.

The same table settles a second term. DA-I, which Younger leaves as a possible total without adding it up, closes HT 12 with 50, which is exactly the sum of the seven entries of the tablet ($5 + 5 + 1 + 30 + 5 + 3 + 1$) if one fraction of $\frac{1}{16}$ is excluded; its second attestation (KH 13) is a tablet-final with a quantity, compatible. DA-I is therefore the third Linear A term whose function is fixed by arithmetic, after KU-RO and KI-RO. For KI-RO, the olive tablet HT 123+124a confirms the reading as deficit: the olive total adds up exactly ($93 \frac{1}{2}$), the product *308 is between 26% and 28% of the olives in each section, and what was delivered plus the final KI-RO gives a third of the total ($25 + 6 \approx 31$); this is the arithmetic Younger performs in his commentary, replicated here independently.

4.6 Form screens and their negative control

A screen of the whole vocabulary against twenty phytonyms attested as commodities in Mycenaean gives 34 stem matches against 15.4 expected (maximum 25 in 200 permutations); against sixty, 91 against 53.1 (maximum 84); against seventy-nine, 97 against 73.7 (maximum 103). The excess concentrates in one family and disappears as the list grows. The same screen by field (animals, vessels, minerals, lists of twenty) stays within chance in all three. Applied to Linear B, it produces 383 matches, most of them with Greek personal names unrelated to plants. Conclusion: a two-syllable match discriminates in neither script, and any reading resting on it is indefensible; what discriminates is the conjunction of stem, distribution and derivation, case by case, as argued in the paper on phytonyms.

4.7 The libation formula: slot function without reading

The eighteen inscriptions carrying at least one anchor of the libation formula allow its template to be fixed by distribution: four nearly invariant anchors and a fixed opening enclose slots whose fillers are singletons in 32 of 32 cases (probability under the corpus singleton rate, of the order of 10^{-4}), so that the slots hold the open class of names and places and the anchors the closed class;

the only inscription with logograms (SY Za 2) fixes the anchor u-na-ka-na-si as the term of the offering without recourse to etymology; three prefixes have measurable distributions; and the template, compared as a structure with contemporary votive formulae, resembles the Egyptian and the Anatolian and not the Semitic or the Greek. Listing the functions each slot admits and discarding those that contradict a datum, two structural readings survive, and the datum that would separate them is not in the corpus. The full analysis, with the candidate sentences and the same treatment of the administrative terms, is presented separately (Acedo 2026f); what matters here is the methodological result: in a corpus without a bilingual one can fix the function of each position of a formula and bound the number of possible readings, not the value of its words.

4.8 A typological profile without a list of candidate languages

The usual question, which family Minoan belongs to, requires choosing a set of comparison languages, and that choice conditions the result: the candidates that would matter (the languages of the pre-Hellenic Aegean, those of western Anatolia before the Hittite texts) are unattested and can enter no list. The answerable question is typological: how the system is built. It is measured without choosing candidates and compared afterwards, if one wishes, with whomever one wishes.

Word length by role in the document. Roundel words average 3.34 syllables ($n = 29$), entries with a quantity 2.81 ($n = 323$), personnel-list entries 2.73 ($n = 11$) and tablet headings 2.68 ($n = 236$); the range between means is 0.67 against 0.31 expected when words are permuted among roles ($p = 0.038$). By known class, toponyms measure 3.43 syllables, the personal names that survive at Knossos 2.90 and the transaction terms 2.25. The gradation (short terms, long names and toponyms) is the same that separates short commodity strings from long name strings in Proto-Elamite, and is a feature of bookkeeping rather than of a language.

Internal predictability of the word. The conditional entropy of a syllable given the previous one is 3.79 bits in Linear A, against a marginal entropy of 5.35: knowing the previous syllable reduces uncertainty by 1.56 bits. In Linear B the reduction is 1.61, and 1.92 when the token count is matched to that of Linear A; in the two Cypriot corpora, much smaller, it is around 2.3, which shows that the measure grows as the corpus shrinks and forces comparison at matched size only. The one usable contrast, Minoan against Greek at equal tokens (1.56 against 1.92), points to a lexicon with less predictable transitions, compatible with short words and without the regular inflectional morphology of Greek. It is an observation, not a proof, and its contamination by size is declared.

Compositionality, and why it is withdrawn. Measuring the proportion of long words whose first two and last two syllabic elements recur elsewhere in the vocabulary, Linear A gives 0.12, against 0.02 for a shuffled-syllable null and against 0.07 for Linear B reduced to the same size and the same lengths; it therefore appeared more compositional than Greek. With the correct null, which preserves syllable transitions (a first-order Markov chain on the corpus itself) as well as lengths, the observed value (0.124) falls within what the null produces (mean 0.131, maximum 0.197; $p = 0.60$): the apparent composition is the phonotactics of the syllabary. The

result is withdrawn. One descriptive fact survives: of the 190 recurrent syllabic elements of the corpus, 142 (75%) are also attested as independent words, against 59% in Linear B.

The methodological lesson is worth more than the withdrawn result, and it is the second of its kind in this paper (the first, the two-sided window of section 5): the null must preserve everything that is not the hypothesis, and in a text written in a syllabary that includes phonotactics. A null of shuffled units is too weak and manufactures positives.

4.9 The affiliation proposals and what each of them predicts

No claim of this work depends on choosing a candidate language, but the reader is entitled to the map of what has been proposed and to know which part of each proposal is testable against the profile of the preceding sections. The table gathers the main hypotheses, the evidence adduced and the objection their own critics raise.

proposal	main proponents	what it rests on	main objection
Semitic (Northwest)	Gordon 1966; Best and Woudhuizen 1989	readings of isolated words (KU-RO as *kull-, “all”); the presence of Semitic loans in the Aegean	the readings do not predict unseen text; Duhoux (1989, 2011) shows that KU-RO requires no Semitic etymology and that the matches are of the order of chance
Anatolian (Luwian or a relative)	Palmer 1958; Finkelberg 1990-91, 2001	toponyms and suffixes in -ss- and -nd-; the libation formula read with a-sa-sa-ra-me ~ Luwian ishassara “lady”	the -ssos and -nthos layer is Greek, not Minoan (see 4.3 bis); the reading of the formula depends on a single word and on an unestablished template
Indo-Iranian or Indo-Aryan	Owens 1996-2007	isolated lexical and morphological correspondences	rejected by the field on method: matches without chance control (Duhoux; Davis 2014)
Kartvelian	Gordeziani 1980-2007	agglutinative typology and lexical comparison	the same problem of control; the agglutinative typology has not been demonstrated in the

proposal	main proponents	what it rests on	main objection
			corpus
Greek	Georgiev 1963 and successors	reading the corpus with Linear B values	discarded by the profile distance (4.1), by the converter (36-47%) and by the absence of Greek morphology
Isolate or lost family	the implicit majority position (Duhoux 1989; Davis 2018; Petrakis and Steele 2025)	absence of systematic correspondences with any attested family	not a testable hypothesis: it is the default state while the others fail to hold

What each predicts about structure, and what we have been able to measure. A Semitic hypothesis predicts discontinuous consonantal roots, a filiation particle or word in the onomastics, and possessive suffixes; the corpus has no filiation word (4.4), which counts against it, but discontinuous roots are not measurable with a syllabary that does not write vowels independently, so the main prediction is inaccessible. An Anatolian hypothesis predicts genitives in -s and adjectives in -ss-, plus a compound onomastics; of those features the corpus allows one to measure the productivity of the suffixes (4.4: -ja, -te, -se above the null, none after correction) and compositionality (4.8: not distinguishable from phonotactics), and the result neither confirms nor refutes. An agglutinative hypothesis, such as the Kartvelian, predicts long words with many chained affixes; the length profile (2.7 to 3.4 syllables, 4.8) is not that of such a language, and that is a measurable point against it. The Greek hypothesis is discarded with numbers (4.1).

The honest conclusion is twofold. First, that the corpus allows only part of what each hypothesis predicts to be measured, and that part does not suffice to choose: two of the proposals (Semitic and Anatolian) make predictions that at this size are indistinguishable from one another. Second, that the features we have measured, the absence of a filiation word, word length, the limited productivity of the affixes and the lack of demonstrable compositionality, form a profile that none of the proposals explains better than the rest. Publishing that profile, together with what each hypothesis would have to predict, is more useful than scoring candidates: it turns a debate of authorities into a list of predictions testable when more text appears.

5. Negative results of method

They are reported because they close paths that others would reopen.

A per-node structural descriptor (Omega-N, with the public code of BiomeMakers/OmegaN) applied to aligning A and B signs by their role in the co-occurrence graph recovers a mean rank of 27.4 against 32.5 for chance ($p = 0.018$), exactly what degree alone recovers (26.8): what transfers between scripts is the degree share, and structural role encodes the language, not the

sign. Within A, the descriptor has no test-retest reliability at these sizes (triadic excess between two halves of Haghia Triada +0.06 and +0.02), so any test of fine structure is undecidable; a minimum-size condition is proposed for the method's screen.

A function classifier (49 labels, 25 features of position and company) scores 0.67 to 0.73 under leave-one-out, against 0.39 to 0.48 with shuffled labels: it distinguishes the libation formula, and headings from entries, but classifies the two known commodities as names, because there are two examples of commodities. Structure predicts coarse function, not semantic class.

A neural decipherment model with and without the Omega-S regulariser on the twenty-five known words is preregistered with a prediction of zero in both arms; it has not been run.

Co-exclusion between words has no power at this size (section 4.5), and the aromatics "recipe" (equal proportions among ingredients) is undecidable with the legible quantities (0.26 of equal pairs against 0.21 for the fraction null, $p = 0.24$).

Three tests taken from network ecology are declared inconclusive or withdrawn. Forman curvature on the co-occurrence graph, with the prediction that transaction terms occupy the bridges, does not separate those terms from the rest at Susa ($p = 0.50$) and at Haghia Triada points in the predicted direction without reaching significance ($p = 0.12$), probably because Forman curvature is dominated by degree; the test requires Ollivier-Ricci curvature and remains pending. The "energy" gradient (documents ordered by volume of commodity) does not alter vocabulary diversity at Susa (singleton rate 0.40, 0.40, 0.36, 0.42 by quartiles; $p = 0.47$), so the analogy with the external energy gradient of soils does not transfer. And Ulanowicz's ascendancy/capacity ratio, which in a first pass placed Haghia Triada and Susa in the window of vitality ($a = 0.41$ and 0.39 , with F maximal at $a = 1/e$), is reproduced exactly by a null preserving the row and column sums of the documents $\times$ words matrix (0.411 and 0.430 in the null), and Uruk and Iberian do not even fall in the window: the index measures the marginal distributions and not the structure, and the result is withdrawn. The lesson is general and holds for any index computed on a count matrix, inside and outside epigraphy: if the fixed-margin null reproduces it, the index measures the margins.

The ligature *304+PA-KU-PA of the LinearA Explorer does not exist: SigLA reads *629 followed by KU-PA. It is reported because it supported for two days an argument for a reading, and because derived transliterations require sign-by-sign verification before use as data.

6. A case: how the procedure produced, and then corrected, a reading

The procedure is best illustrated by the case in which it half failed. Searching for the sign-groups that behave as counted things (recurrence within a document with different quantities, integer quantities, absence from personnel lists), the screen isolated KU-PA (AB 81-03): repeated on both faces of ZA 11 with 1 and 3, on KH 29 with 2, on HT 110a with 1; with a family of derivatives (KA-KU-PA, KU-PA-ZU, KU-PA-JA, KU-PA-RI) that stand beside the logogram *303 on two roundels and on KH 5; and with two affixes that recur in the corpus above the null. Only then was the form compared: Greek κύπαιρος, Pre-Greek according to Frisk and Beekes,

Mycenaean ku-pa-ro, which glosses the cyperus logogram on KN Ga 517. The first version read the family as the Minoan word for cyperus, against Younger's reading, which took it for the name of a contributor.

Three later checks, all from the same procedure, changed the reading without touching the etymology. Sign-by-sign verification in SigLA showed that the "ligature" *304+PA-KU-PA on the nodule HT Wa 1020, one of the supports, was the logogram *629 followed by two signs. Tabulating the seals of all the Haghia Triada roundels (Younger's commentaries, SigLA) showed two sealing groups: one in which the word stands alone, and another in which the word is followed by a commodity logogram, and in that second group stand KA-KU-PA with *303 but also A-RA-TU-ME with ewes and KU-MI-NA-QE with goats, words that cannot be commodities. And the structure of KH 5 (group, logogram, quantity, with WI-SA-SA-NE before *626 and KU-PA-ZU before *303) turned out to be that of parties before commodities. The corrected reading is the one that reconciles: the stem KU-PA is the plant's word; the family in the archives is a name formed on it, like Kyparissos or Marathon in Greek; and two further cases of the same type (KU-MI-NA-QE and cumin; KI-KI-RA-JA on the KI-KI-NA that Neumann identified with the sycamore fig in 1960) sustain the pattern.

What the case teaches about the method is what matters here. First, that the function-before-form order does not by itself protect against an error of function: recurrence with quantities, taken as the signature of a commodity, was compatible with a party allotted twice, and only a datum external to the tablet (the seals) decided it. Second, that every support must be verified in the primary source and not in a derived transliteration, however good. Third, that the correction did not destroy the result but placed it: the etymology, the only thing that did not depend on function, survived intact, and the pattern of names formed on plants is more defensible and more general than the initial reading. The paper presenting the case (Acedo 2026, under evaluation) carries that history inside it because the history is part of the evidence.

7. Discussion

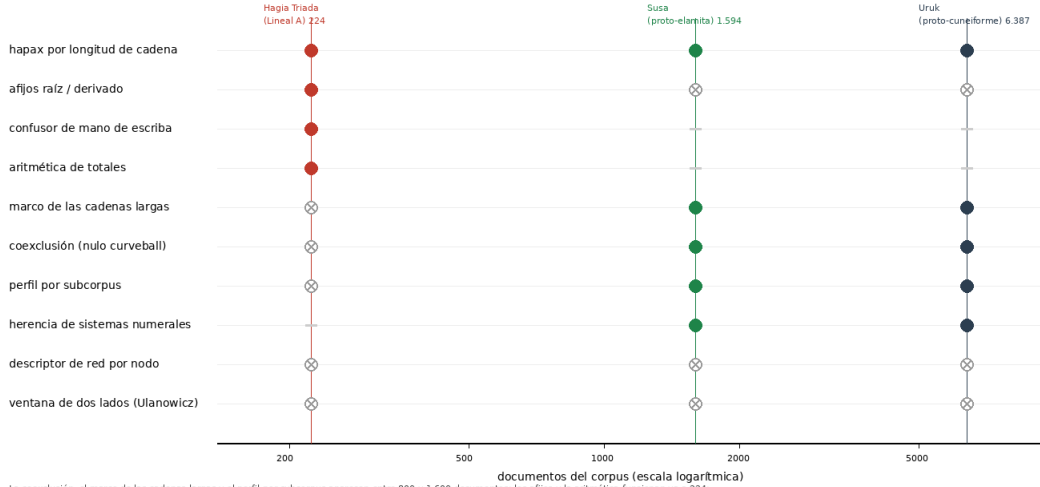
The procedure does not decipher, and it is worth saying so with the figures of this paper: on 7,000 signs it produced two stems with a proposed meaning, two names decomposed, two affixes with a placed function, and the template of one formula. What it produced in greater abundance are negatives with numbers, and that is its value: each closes a path that the literature has reopened for a century. Comparison by sound does not discriminate among candidate languages (all at 1.3-1.7 over their null), neither in Linear A nor in Linear B (383 false matches under the same rule); the difference between archives and sanctuaries is not one of administrative type; co-exclusion is not measurable at 224 samples; a network descriptor is not reliable at this size; a classifier does not learn a class from two examples.

That the negatives are of size and not of method is shown by cross-calibration. The same apparatus, on Proto-Elamite (1,594 tablets, 8,282 entries), recovers in one session the classification of signs by numeral system, the institutional header and the name/commodity partition that the field established over twenty years, and finds in addition the frame of the long

strings, 24 grain monopolies and 13 co-exclusion pairs under the curveball null, which in Linear A had no power. On Uruk (6,387 tablets), it recovers three of the four predictions on the inheritance of numeral systems and refutes the fourth by absence. On Eteocypriot, with 21 inscriptions, the profile test is declared without power and the positional test separates the two languages by their endings ($p < 10^{-16}$). On Iberian, with 1,919 texts and only the province as metadata, it recovers Untermann's onomastic formants (295 against 11 for the null) and the two known grapho-morphological isoglosses (the final southern sign S56, $p = 3 \cdot 10^{-9}$; the north-eastern suffix *-mi*, $p = 3 \cdot 10^{-5}$). On Cypro-Minoan, where there is no known answer to calibrate against, the same apparatus finds structure where the null does not produce it: in the 183 inscriptions of the dataset published by Corazza, Tamburini, Valério and Ferrara (2022), sequences of three and four signs recurring at least three times number 19 and 6, against 0.9 and 0 when signs are shuffled within each inscription preserving lengths and frequencies ($p < 0.005$), so that the script combines signs into recurrent units; the corpus does not, on the other hand, allow the archive analysis, since only six of its inscriptions are tablets. In every corpus the instrument says the same thing: what there is, what there is not, and what cannot be said with that n.

A qué tamaño de corpus empieza a ver cada instrumento

circulo lleno: detecta la estructura · aspa: no la detecta, o el nulo de márgenes la reproduce · raya: no aplicable a ese corpus



La coexclusión, el marco de las cadenas largas y el perfil por subcorpus aparecen entre 800 y 1.600 documentos; los afijos y la aritmética funcionan ya a 224; el control de mano solo donde hay atribución de escribas; el descriptor por nodo y la ventana de dos lados no ven en ningún tamaño de los probados.

Fig. 1. At what corpus size each instrument begins to see. Filled circle: it detects the structure; cross: it does not, or the fixed-margin null reproduces it; dash: not applicable. Co-exclusion, the frame of long strings and the profile by subcorpus appear between 800 and 1,600 documents; affixes and arithmetic already work at 224; the hand control only where scribes are attributed.

A note on procedure, so that the figures above are read correctly. The work had an exploratory phase, without prior hypotheses, from which nothing is asserted on its own: its results were treated as hypotheses and only those replicated with a specific null or confirmed in another corpus are retained. The new claims come from the preregistered phase, in which each hypothesis carried its null, its threshold and its stopping rule before being run. Within each

family of tests, multiple comparisons are corrected (Bonferroni on the eleven suffixes, where none survives the corrected threshold, and on the sixty Iberian endings; Benjamini-Hochberg on the sign-and-system associations), and tests of a single preregistered hypothesis are reported with their exact p. Three results of the exploratory phase did not survive and are declared withdrawn in section 5.

The origin of the apparatus is community ecology, and the analogy deserves one sentence, no more: documents are samples, words are taxa, and the questions about co-occurrence, co-exclusion, specificity and turnover are the same as those asked of a microbial community, with the same nulls (Gotelli 2000; Strona et al. 2014) and with the same test-retest reliability requirement that any index of structure demands (Acedo 2026b). What epigraphy adds to ecology is a confounder that in biological samples is called batch and here is called scribe: at Haghia Triada shared vocabulary is explained by the hand and not by the document type ($p = 0.002$ against 0.21), and a niche index that did not control for it would be measuring scribal portfolios. What ecology adds to epigraphy is the habit of asking, before reading, whether the corpus has the size to answer.

Where this leads is not towards reading. It is towards anchors: every Pre-Greek word whose stem appears in the tablets with the right function is one line of a synthetic bilingual, and with thirty such lines Minoan would have a vocabulary, a morphology on that vocabulary and a formula with parts, even if it were not pronounced. The procedure described here is the factory of those anchors, and the guarantee that each carries its null, its calibration and its correction when due.

Declaration of assistance

Corpus processing, permutation tests and the drafts of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

Appendix: equations

- (1) Jensen-Shannon distance: $JSD(P,Q) = \frac{1}{2} D(P\|M) + \frac{1}{2} D(Q\|M)$, $M = (P+Q)/2$, D the Kullback-Leibler divergence in base 2 (Lin 1991).
- (2) Jaccard index: $J(A,B) = |A \cap B| / |A \cup B|$ (Jaccard 1901).
- (3) Mantel correlation between distance matrices, with row-and-column permutation (Mantel 1967).
- (4) Mutual information between positions: $I(X;Y) = \sum p(x,y) \log_2 [p(x,y) / p(x)p(y)]$ (Shannon 1948).
- (5) Fixed-margin null by the curveball algorithm (Strona et al. 2014), which preserves the row and column sums of the documents $\times$ words matrix.

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